

Properties of Integration [Arithmetic]

Let a, b and A, B, C be real numbers. Let the functions $f(x)$ and $g(x)$ be integrable on an interval that contains a and b . Then

$$1. \int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

$$2. \int_a^b [C f(x)] dx = C \int_a^b f(x) dx$$

$$3. \int_a^b [A f(x) \pm B g(x)] dx = A \int_a^b f(x) dx \pm B \int_a^b g(x) dx$$

$$4. \int_a^a f(x) dx = 0$$

$$5. \int_b^a f(x) dx = - \int_a^b f(x) dx$$

$$6. \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx ; a \leq c \leq b$$

7. If $f(x)$ is an even function, then

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx, a > 0$$

8. If $f(x)$ is an odd function, then

$$\int_{-a}^a f(x) dx = 0$$

Antiderivative :

A function F is called an antiderivative of the function f on a given interval I if $F'(x) = f(x) \quad \forall x \in I$.

So,

$$F'(x) = f(x)$$

$$\frac{dF(x)}{dx} = f(x)$$

$$dF(x) = f(x) dx$$

$$\boxed{\cancel{F(x)} = \int \cancel{f(x) dx}}$$

$$\int dF(x) = \int f(x) dx$$

$$\boxed{\int f(x) dx = F(x) + c}$$

↳ antiderivative

* previous discussion we used

$$A'(x) = f(x)$$

both are same.

e.g. $F(x) = \frac{x^3}{3}$ is ^{an} antiderivative of $f(x) = x^2$

Similarly, $F(x) = \frac{x^3}{3} + 2$ or $\frac{x^3}{3} - 5$ are also antiderivative of $f(x) = x^2$.

which means $\cancel{f(x)}$ $\cancel{F(x)}$ is also an antiderivative of $f(x) = x^2$.

Thus $\int f(x) dx = F(x) + c$, c is called integral constant

$$\text{or. } \int x^2 dx = \frac{x^3}{3} + c$$

$F(x) + c$ is called family of curve !!!

Comp

Theorem: If f is integrable on $[a, b]$ and $f \geq 0$, then

$$\int_a^b f(x) dx \geq 0.$$

Comparison Theorem: Let f and g be integrable on $[a, b]$ and suppose that for every x in $[a, b]$, $f(x) \leq g(x)$.

Then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

Proof: Define $h(x) = g(x) - f(x)$.

Given that $f(x) \leq g(x)$ which implies $h(x) \geq 0$ on $[a, b]$.

Since f and g be integrable on $[a, b]$ so $h(x)$ is also integrable on $[a, b]$. By above theorem,

$$\int_a^b h(x) dx \geq 0.$$

$$\begin{aligned} \text{But } 0 \leq \int_a^b h(x) dx &= \int_a^b [g(x) - f(x)] dx \\ &= \int_a^b g(x) dx + \int_a^b [-f(x)] dx \\ &= \int_a^b g(x) dx - \int_a^b f(x) dx \end{aligned}$$

$$\text{Therefore } \int_a^b f(x) dx \leq \int_a^b g(x) dx$$



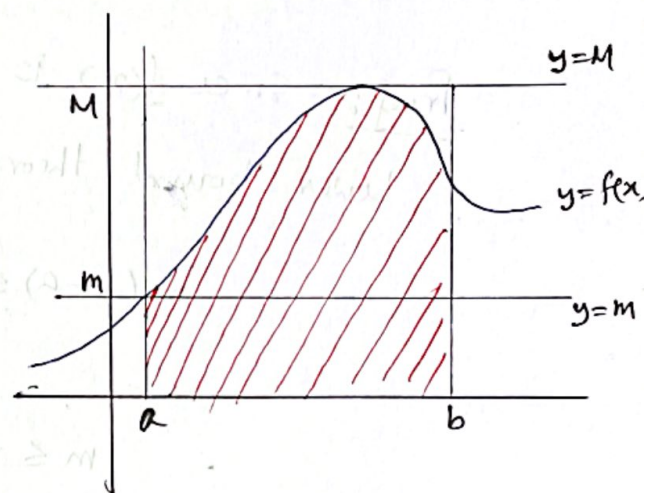
Upper and Lower Bound Theorem

Suppose that on $[a, b]$

$$m \leq f(x) \leq M \quad \text{for all } x \text{ in } [a, b].$$

That is, m and M are lower and upper bounds, respectively, for the function f on the interval $[a, b]$. Then if f is integrable on $[a, b]$,

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a).$$



PROOF: If we define g by

$$g(x) = M,$$

then $f(x) \leq g(x)$ on $[a, b]$

By Comparison Theorem,

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx = \int_a^b M dx = M(b-a)$$

Similarly,
$$m(b-a) = \int_a^b m dx \leq \int_a^b f(x) dx$$

and the theorem is proved. $\square$

The Mean Value Theorem for Integrals

If $f(x)$ is continuous over an interval $[a, b]$, then there is at least one point $c \in [a, b]$, such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx$$

OR

$$\int_a^b f(x) dx = f(c)(b-a).$$

Proof: Since $f(x)$ is continuous on $[a, b]$. By the upper and lower bound theorem

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

$$m \leq \frac{1}{b-a} \int_a^b f(x) dx \leq M$$

Since $\frac{1}{b-a} \int_a^b f(x) dx$ is a number between m and M , and since $f(x)$ is continuous and assumes the values of m and M over $[a, b]$. By the IVT, there is a number $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

□

Problem: Show that the value of $\int_0^1 \sqrt{1+\cos x} \, dx$ is less than or equal to $\sqrt{2}$.

Sol: By upper and lower bound Theorem,

$$\text{say, } m \leq f(x) \leq M$$

$$\text{Here } f(x) = \sqrt{1+\cos x}$$

$$\text{so, } m(b-a) \leq \int_a^b f(x) \, dx \leq M(b-a)$$

$$M = \text{Upper bound of } f(x) = \text{Max } f(x) = \text{Max } \sqrt{1+\cos x} \text{ on } [0,1] \\ = \sqrt{2}$$

Since Maximum value of $\sqrt{1+\cos x}$ on $[0,1]$ is $\sqrt{1+1} = \sqrt{2}$

$$\text{so, } \int_0^1 \sqrt{1+\cos x} \, dx \leq \sqrt{2} \cdot (1-0) = \sqrt{2}$$

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EXTRA PROBLEMS:

Book: THOMAS'

PAGE: 327-328

PROBLEMS: 71, 72, 73, 74, 75, 76.

PROBLEM: Let $f(x) = x^2 - 2x + 1$ on $[0, 4]$

(i) Find the min and max values on the interval.

(ii) Use both bounds to estimate the integral.

(iii) Check the exact integral value.

Soln:

Here $f(x) = x^2 - 2x + 1 = (x-1)^2$ on $[0, 4]$

Since $f(x) \geq 0 \forall x$

i.e. $(x-1)^2 \geq 0$

$m = \min f(x) = 0$ when $x=1$

$M = \max f(x) = 9$

$$f(0) = (0-1)^2 = 1$$

$$f(4) = (4-1)^2 = 3^2 = 9$$

Now, Upper and lower bound theorem,

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

$$0 \cdot (4-0) \leq \int_0^4 (x^2 - 2x + 1) dx \leq 9 \cdot (4-0)$$

$$0 \leq \int_0^4 (x^2 - 2x + 1) dx \leq 36$$

For exact:

$$\begin{aligned} \int_0^4 (x^2 - 2x + 1) dx &= \left[\frac{x^3}{3} - x^2 + x \right]_0^4 = \frac{1}{3} 64 - 16 + 4 = \frac{28}{3} \\ &= \frac{28}{3} \approx 9.333. \end{aligned}$$

Fundamental Theorem of Calculus : Part I

If f is continuous over an interval $[a, b]$, and the function $F(x)$ is defined by

$$F(x) = \int_a^x f(t) dt,$$

then $F'(x) = f(x)$ over $[a, b]$.

Proof: Here, we have

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\int_a^{x+h} f(t) dt - \int_a^x f(t) dt \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\int_a^{x+h} f(t) dt + \int_x^a f(t) dt \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t) dt \end{aligned}$$

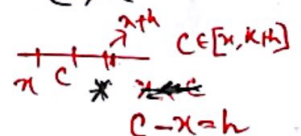
By the MVT, there is some c in $[x, x+h]$ such that

$$\frac{1}{h} \int_x^{x+h} f(x) dx = f(c)$$

Since $c \in [x, x+h]$ and f is continuous, so, $\lim_{h \rightarrow 0} f(c) = \lim_{c \rightarrow x} f(c) = f(x)$

$$\text{Therefore } F'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(x) dx$$

$$= \lim_{h \rightarrow 0} f(c) = f(x)$$



The Fundamental Theorem of Calculus : Part 2

If f is continuous over the interval $[a, b]$ and $F(x)$ is any antiderivative of $f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a) .$$

Proof: let $P = \{x_i\}$, $i = 0, 1, \dots, n$ be a regular partition of $[a, b]$. Then, we can write

$$\begin{aligned} F(b) - F(a) &= F(x_n) - F(x_0) \\ &= F(x_n) - F(x_{n-1}) + F(x_{n-1}) - F(x_{n-2}) + \dots \\ &\quad + F(x_1) - F(x_0) \\ &= [F(x_n) - F(x_{n-1})] + [F(x_{n-1}) - F(x_{n-2})] + \dots \\ &\quad + [F(x_1) - F(x_0)] \\ &= \sum_{i=1}^n [F(x_i) - F(x_{i-1})] \end{aligned}$$

By the MVT, for $i = 0, 1, \dots, n$ we can find c_i in $[x_{i-1}, x_i]$ such that

$$\begin{aligned} F(x_i) - F(x_{i-1}) &= F'(c_i) (x_i - x_{i-1}) \\ &= f(c_i) \Delta x \end{aligned}$$

$$\begin{aligned} \text{So, } F(b) - F(a) &= \sum_{i=1}^n f(c_i) \Delta x \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x \\ &= \int_a^b f(x) dx \end{aligned}$$



Problem: Given $f(t) = \int_0^t \frac{\tilde{x}^2 + 8\tilde{x} + 12}{1 + \cos \tilde{x}} d\tilde{x}$. At what value of t does the local max of $f(t)$ occur?

Solⁿ: Fundamental Theorem of Calculus,

$$f'(t) = \frac{t^2 + 8t + 12}{1 + \cos t}$$

Since $1 + \cos t > 0$, so $t^2 + 8t + 12 = 0$ for relative max/min of $f(t)$.

$$\text{Now, } t^2 + 8t + 12 = 0$$

$$t^2 + 6t + 2t + 12 = 0$$

$$(t+6)(t+2) = 0$$

$$t = -2, -6$$

when $t < -6$ $f'(t) > 0$ so, $f \uparrow$

when $-6 < t < -2$ $f'(t) < 0$ so, $f \downarrow$

when $t > -2$ $f'(t) > 0$ so, $f \uparrow$

Thus f has a local maximum at $t = -6$ and local minimum at $t = -2$.

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